

Loan Default Prediction with Explainable AI

Project Overview

Overview

Loan default prediction is a critical application in the financial industry, where lenders and institutions aim to assess the risk associated with providing loans to individuals or businesses. The goal is to identify potential borrowers more likely to default on their loans, allowing lenders to make informed decisions and mitigate financial risks.

In the context of loan default prediction, machine learning models have shown promising results in accurately predicting whether a borrower will default on a loan. These models leverage historical data, such as past loan performance, financial history, employment details, and other relevant factors, to predict the likelihood of future loan defaults.

However, as machine learning models become increasingly complex, a significant challenge arises: the lack of transparency. Many modern machine learning algorithms, such as deep neural networks, can be seen as "black boxes." While they may produce accurate predictions, how they arrive at those predictions is often unclear. This opacity raises concerns for various stakeholders involved in the loan lending process:

- **Data Scientists:** Data scientists who develop and deploy these models need to understand their inner workings for debugging, improvement, and model selection. They require insights into feature importance and decision processes to ensure models are reliable and accurate.
- **Stakeholders:** Business stakeholders, such as loan officers and management teams, need to trust and comprehend the model's predictions to make well-informed lending decisions. Clear explanations are essential for building trust and facilitating collaboration between data scientists and business units.
- **Regulators:** Regulatory bodies and compliance officers demand transparency in lending practices. They require models to be explainable and auditable to ensure fairness, ethical considerations, and compliance with regulations like anti-discrimination laws.

Aim

This project aims to predict loan defaults using machine learning models (XGBoost and Random Forest) and then enhance model interpretability using Explainable AI techniques. The project seeks to clearly understand why the models make certain predictions.

Data Description

The dataset comprises attributes such as user IDs, loan categories, amounts, interest rates, tenure, employment details, income, and more. The target variable is "Defaulter," indicating whether a loan will default.

Tech Stack

- Language: Python 3.10.4
- Libraries: pandas, numpy, Graphviz, scikit-learn, Neptune, XGBoost, Hyperopt, Deepchecks, SHAP

Approach

- Part 1: Data Preparation and Model Building
 - Data Preparation:
 - Load and merge Excel sheets.
 - Explore data statistics, distributions, and data types.
 - Handle missing values using appropriate techniques.
 - Address data skewness and multicollinearity.
 - Encode categorical features.
 - Address data imbalance for loan defaults.
 - Model Building and Tuning:
 - Split the data into training and testing sets.
 - Train XGBoost and Random Forest models.
 - Utilize Neptune for experiment tracking.
 - Fine-tune models using Hyperopt and Grid Search.
- Part 2: Explainable AI
 - Overall Model Performance Analysis:
 - Evaluate classification metrics and confusion matrix.
 - Understand recall, precision, and F1 score.
 - Discuss the metric impact on stakeholders.
 - Deepchecks for Model Enhancement:
 - Utilize Deepchecks to identify model weaknesses.

- Suggest strategies to improve model performance.
- Exploring SHAP (Shapley Additive exPlanations):
 - Use SHAP for feature importance explanations.
 - Create SHAP dependence plots for subsegments.
 - Visualize SHAP values for individual predictions.
- Introducing Anchors for Intuitive Explanations:
 - Implement Anchors for human-understandable explanations.
- Counterfactuals for Business Insights:
 - Explore counterfactual explanations for feature impacts.
 - Tailor interpretations for stakeholders and regulators.

Modular code overview:

Once you unzip the modular_code.zip file, you can find the following folders.

```

├─ engine.py
├─ Input
│   ├── CleanedData.xlsx
│   ├── Cleaned_SampledData.xlsx
│   └── Credit_Risk_Dataset.xlsx
├─ Notebooks
│   ├── 1. Data prep and ML.ipynb
│   ├── 2. Explainable AI.ipynb
│   └── Source
├─ Output
├─ readme.md
├─ requirements.txt
└─ Source
    ├── eda.py
    ├── machine_learning.py
    └── setup.py
  
```

1. Input
2. Output
3. notebooks

4. Source
 5. engine.py
 6. requirements.txt
 7. readme.md
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1. The notebook folder is a reference folder and contains the original ipython notebook as in the lectures.
 2. The source folder contains all the functions put into different python files, which are appropriately named. The engine.py script then calls these python functions to run the steps in one go to train the model and print the results.
 3. The requirements.txt file has all the required libraries with respective versions. Kindly install the file using the command **pip install -r requirements.txt**
 4. **All the instructions for running the code are present in readme.md**

Project Takeaways

1. Understanding the significance of loan default prediction and its impact on financial decision-making
2. Understand the data preprocessing techniques to clean, encode, and balance data for accurate model training
3. Utilizing XGBoost and Random Forest models for predicting loan defaults.
4. Hands-on experience in Hyperopt and Grid Search for fine-tuning model hyperparameters
5. Learn Neptune for experiment tracking, ensuring organized and traceable model development
6. Interpretation of key classification metrics like accuracy, precision, recall, and F1 score for model evaluation
7. Insight into confusion matrix analysis, aiding in understanding true positive and false negative rates
8. Awareness of the implications of metrics on stakeholders and their decision-making processes
9. Practical implementation of Deepchecks to identify areas for model improvement, leading to enhanced accuracy

10. Explaining feature importance using SHAP, identifying factors influencing loan default predictions
11. Creation of SHAP dependence plots to understand feature interactions and their impact on predictions
12. Understanding individual prediction explanations through SHAP values, enhancing model transparency
13. Application of Anchors for intuitive explanations, making complex model decisions understandable
14. Exploration of counterfactual explanations for understanding how feature changes impact loan approval
15. Realization of the ethical and regulatory aspects of AI models, ensuring fairness and compliance
16. Enhanced communication skills to convey technical concepts to non-technical stakeholders effectively